
Graph-Based Stackelberg Model for Adversarial Supply Chain Enforcement

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Abstract

Advanced semiconductor chips underpin modern artificial intelligence systems and have become a strategic driver of military, economic, and technological power. As access to high-end computing increasingly shapes national security outcomes, the United States and allied governments have expanded export controls on advanced computing chips and semiconductor manufacturing equipment. The effectiveness of these controls, however, depends critically on enforcement capacity. Enforcement agencies operate under severe resource constraints relative to the scale and complexity of global semiconductor supply chains. At the same time, smuggling networks can rapidly reconfigure intermediaries, routing, and documentation across permissive jurisdictions. This mismatch creates a structural enforcement challenge. Agencies must commit to inspection and monitoring strategies before observing how adversaries route shipments; adversaries can observe and adapt to these commitments, and enforcement resources are strictly limited relative to the number of potential targets. These features naturally give rise to a leader-follower interaction between regulators and smuggling networks. In this paper, we formalize semiconductor diversion as a Stackelberg security game on a directed supply chain graph. An enforcement agency commits to a probabilistic inspection policy, and an adversary subsequently adapts routing decisions to maximize illicit supplier to end user throughput subject to capacity and budget constraints. The resulting model captures the strategic commitment, adaptive behavior, and resource asymmetries that characterize real-world export control enforcement. By framing enforcement as a network interdiction problem with stochastic detection and constrained evasion, the model provides a foundation for analyzing interdiction effectiveness under adaptive adversarial pressure and limited enforcement capacity.

1 Introduction and Motivation

Semiconductor chips provide the computational foundation for modern artificial intelligence systems. The training and deployment of large-scale models in language, vision, and autonomy rely on advanced accelerator hardware, particularly high-end graphics processing units and specialized AI chips [[Semiconductor Industry Association, 2025](#)]. Beyond AI, semiconductors are critical inputs to defense systems, telecommunications infrastructure, medical devices, and energy networks, and have been designated as essential across all sixteen U.S. critical infrastructure sectors [[Executive Office of the President, 2026](#)]. As AI capabilities become increasingly central to military and economic power, access to advanced semiconductor technology has emerged as a core national security concern.

Given their strategic importance, governments restrict the sale of the most advanced chips to certain foreign buyers through export control regimes. In the United States, these controls are administered by the Bureau of Industry and Security (BIS) within the Department of Commerce. Since October 2022, BIS has issued multiple rounds of controls targeting advanced computing chips, semiconductor manufacturing equipment, and related technologies, with parallel measures adopted by allied governments in Japan and the Netherlands [[Bureau of Industry and Security, 2022, 2023, Sutter, 2025](#)]. The stated objective of these

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controls is to prevent adversarial states from acquiring computational capabilities that could accelerate military modernization, weapons development, or mass surveillance.

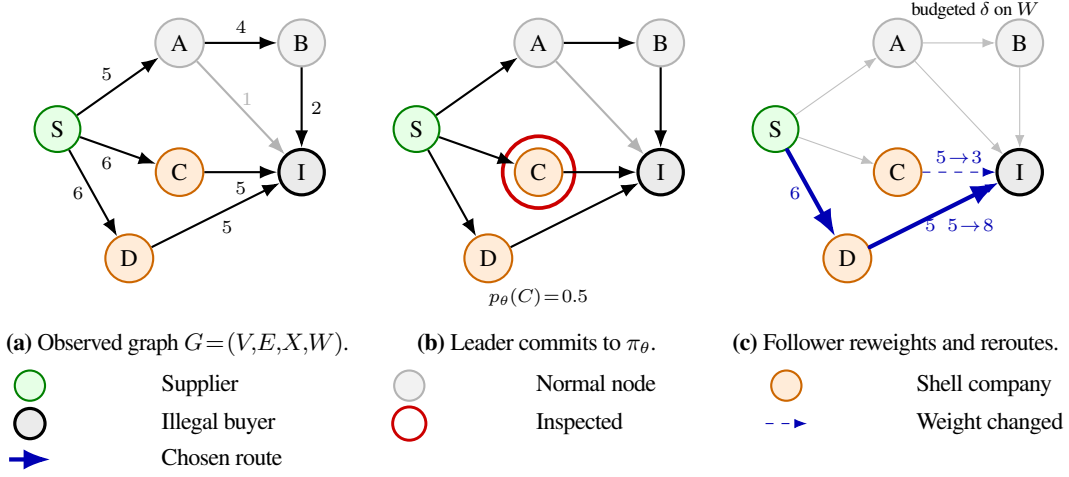


Figure 1: Three-panel schematic of the graph-based Stackelberg enforcement setting: (a) observed directed supply-chain network with baseline edge capacities; (b) the enforcement agency (leader) commits to a probabilistic node inspection policy π_θ ; (c) the smuggling network (follower) adapts by selecting a budgeted edge-weight perturbation δ , yielding $W_f = W + \delta$, and reroutes flow accordingly, illustrating substitution toward an uninspected shell company following partial interdiction.

The effectiveness of export controls, however, depends not only on legal restrictions but on enforcement capacity. Enforcement agencies face a pronounced resource asymmetry. While the global semiconductor trade exceeded \$630 billion in 2024, BIS operated on a budget of approximately \$191 million and maintained a limited number of field agents responsible for monitoring compliance across multiple countries [Sutter, 2025, Grunewald and Fist, 2025]. Individual agents may oversee hundreds of entities and transactions, creating a substantial gap between regulatory scope and enforcement capacity.

Recent enforcement actions illustrate both the scale of semiconductor diversion and the methods used to evade controls. A common strategy involves the creation of shell companies that purchase controlled chips from authorized distributors and reroute them to prohibited end users through transshipment hubs. In 2024, a shell company purchased approximately \$390 million in servers containing restricted Nvidia GPUs before the scheme was detected [Shivakumar et al., 2025]. In another case, a Houston-based firm acquired over \$160 million in advanced GPUs and attempted to route them to China using falsified shipping documents and payments funneled through accounts in Southeast Asia [U.S. Department of Justice, U.S. Attorney’s Office, Southern District of Texas, 2025]. Other networks have used real estate firms or logistics intermediaries as fronts for diversion [U.S. Department of Justice, U.S. Attorney’s Office, Middle District of Florida, 2025]. Syntheses of investigative reporting suggest that smuggled chips may account for a nontrivial share of China’s AI inference capacity acquired in 2024, although precise estimates remain uncertain [Grunewald and Fist, 2025].

Existing enforcement mechanisms struggle to keep pace with these practices. Rule-based screening relies on known red flags and entity lists, but can be circumvented by routing transactions through jurisdictions with limited information sharing or by embedding controlled chips within larger systems that fall outside narrow regulatory definitions [Bureau of Industry and Security, 2025]. Manual investigations are a resource intensive and slow relative to the speed with which shell companies can be created and dissolved. Reactive designation mechanisms such as the Entity List operate only after violations are confirmed, by which point intermediary firms may already have been replaced. Across these approaches, the core challenge is not a lack of detection techniques, but a mismatch between the speed and adaptability of adversarial networks and the limited, pre-committed resources available to enforcement agencies.

This enforcement problem exhibits three structural features that point toward a game-theoretic formulation. First, enforcement agencies must commit to inspection and monitoring strategies before observing how adversaries route shipments, because enforcement posture is partially observable through public actions and disclosures. Second, adversaries can observe these commitments and adapt their routing strategies

in response. Third, enforcement resources are strictly limited relative to the number of potential targets. This structure corresponds to a Stackelberg security game, in which a defender commits to a strategy, and an attacker best responds [Tambe, 2012, Conitzer and Sandholm, 2006]

Main contributions. Motivated by these structural features, we formalize a modeling framework for enforcement resource allocation in adversarial semiconductor supply chains. We represent technology diversion as a Stackelberg security game on a directed supply-chain graph, in which an enforcement agency commits to a probabilistic inspection strategy and an adversary subsequently adapts routing decisions subject to capacity and budget constraints. This formulation captures the structural asymmetries, commitment requirements, and adaptive behavior that characterize real-world export-control enforcement, and provides a foundation for analyzing interdiction effectiveness under resource scarcity.

2 Problem Formulation

Supply chain graphs. We assume the existence of a weighted supply-chain graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, X, W, Y)$, where $\mathcal{V}$ is a set of nodes (e.g., firms or facilities), $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is a set of edges encoding supplier–customer relations, $X \in \mathbb{R}^{|\mathcal{V}| \times d}$ is a matrix of real-valued node attributes, and $W : \mathcal{E} \rightarrow \mathbb{R}_{\geq 0}$ assigns a nonnegative weight to each edge (e.g., transaction volume, dependency strength, or shipment frequency). Each node $v \in \mathcal{V}$ is associated with an attribute vector $\mathbf{x}_v \in \mathbb{R}^d$ from a common feature space. In addition, each node has a categorical label $y_v \in \{0, 1, 2, 3, 4\}$ indicating its role in the supply chain: 0 (supplier), 1 (normal intermediate seller), 2 (shell company), 3 (normal buyer), and 4 (illegal buyer). We collect node labels into a vector $Y \in \{0, 1, 2, 3, 4\}^{|\mathcal{V}|}$ which are then represented in a one-hot manner. To crystallize our problem definition later, we define the supplier set $\mathcal{S} = \{v \in \mathcal{V} : y_v = 0\}$, malicious shell company set $\mathcal{M} = \{v \in \mathcal{V} : y_v = 2\}$, and illegal-buyer set $\mathcal{I} = \{v \in \mathcal{V} : y_v = 4\}$.

When convenient, we equivalently represent the graph $\mathcal{G} = (\mathcal{V}, A, X, Y)$, where $A \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$ is a (possibly asymmetric) weighted adjacency matrix with entries $A_{uv} = W(u, v)$ if $(u, v) \in \mathcal{E}$ and $A_{uv} = 0$ otherwise. We assume $\mathcal{G}$ contains no self-loops, i.e., $A_{vv} = 0$ for all $v \in \mathcal{V}$. Unless otherwise stated, edges are directed to reflect the orientation of the underlying supply relation, but the same notation applies to the undirected case by taking A symmetric.

2.1 The Supply Chain Interdiction Problem

Stackelberg security games over a graph. A Stackelberg security game (SSG) models a defender who *commits* to a (possibly randomized) resource-allocation strategy, after which an attacker *observes* this commitment and chooses a best-response action. Payoffs depend on whether adversarial activity occurs at *covered* (inspected) versus *uncovered* (uninspected) targets [Tambe, 2012, Conitzer and Sandholm, 2006]. In our setting, targets are nodes of a *known and observed* supply-chain graph $\mathcal{G}$, and defender resources correspond to limited inspections of a node by an enforcement agency. Moreover, the attacker best responds by manipulating edge weights to increase illicit supplier-to-illegal-buyer throughput subject to a fixed budget to avoid trivial and unrealistic scenarios —e.g., setting an edge weight to $+\infty$ between a shell company and the illegal buyer, which acts as a sink.

Leader commitment and inspection coverage. To motivate actual practice, an enforcement agency acts as the leader and commits to an inspection policy that seeks to uncover malicious shell companies that act as proxy gateways to siphon chips into sanctioned markets through sell-offs to illegal buyers. We model this by denoting the leader (defender) commitment to a parametric inspection policy:

$$\pi_\theta : (W, X) \mapsto p_\theta \in \Delta^{n-1}, \quad (1)$$

which outputs an inspection distribution $p_\theta(v)$ over nodes with Δ^{n-1} being the $n - 1$ dimensional probability simplex. This distribution is the leader’s belief based on the graph topology and features in $\mathcal{G}$ that certain nodes $v \in \mathcal{V}$ need an allocation of resources, i.e., interdiction. Given a realized graph, the leader allocates a budget of N inspections by sampling an inspection set $I \subset \mathcal{V}$, by drawing *without replacement* according to π_θ . The resulting *marginal coverage (inclusion) probability* of node v belonging to the inspection set is denoted:

$$c_{\pi_\theta}(v) = \mathbb{P}(v \in I | \pi_\theta), \quad \forall v \in \mathcal{V}. \quad (2)$$

Since exactly N distinct nodes are inspected, these inclusion probabilities satisfy the consistency condition

$$\sum_{v \in \mathcal{V}} c_{\pi_\theta}(v) = \sum_{v \in \mathcal{V}} \mathbb{E}[\mathbf{1}(v \in I)] = \mathbb{E}[|I|] = N. \quad (3)$$

Intuitively, once the enforcement agency commits to a policy, we observe the outcome—i.e., the interdicted nodes. While this simplifies the problem for the adversary, in that the adversary has total knowledge of the interdiction states, it is still a meaningfully difficult optimization problem. Furthermore, it is possible to also consider the more challenging case of only observing π_θ and not the outcomes, which adds technical complexity. For the purposes of this paper, we restrict our attention to the simpler and more intuitive case.

Follower attack model. After observing π_θ , the follower (adversary) perturbs edge weights by choosing $\delta \in \mathbb{R}^{n \times n}$, yielding $\widetilde{W} = W + \delta$, subject to the feasible attack set

$$\mathcal{D}(W) = \left\{ \delta : \delta_{uv} = 0 \text{ if } (u,v) \notin \mathcal{E}; |\delta_{uv}| \leq \Gamma \forall (u,v) \in \widetilde{\mathcal{E}}; \sum_{(u,v) \in \widetilde{\mathcal{E}}} |\delta_{uv}| \leq B; W + \delta \geq 0 \right\}. \quad (4)$$

Here $\widetilde{\mathcal{E}} := \bigcup_{v \in \mathcal{M}} \mathcal{N}_1(v)$ is the edge set belonging to all nodes that fall in the one-hop neighborhood, $\mathcal{N}_1$, of any shell company $v \in \mathcal{M}$. In addition, Γ bounds the per-edge perturbation magnitude, and B is the total attack budget. Given capacities $\widetilde{W} \in \mathbb{R}_{\geq 0}^{n \times n}$, define the feasible set of flows on the original edge set $\mathcal{E}$ as

$$\mathcal{F}(\widetilde{W}) = \left\{ f \in \mathbb{R}_{\geq 0}^{n \times n} : f_{uv} = 0 \text{ if } (u,v) \notin \mathcal{E}; \sum_{u:(u,v) \in \mathcal{E}} f_{uv} = \sum_{w:(v,w) \in \mathcal{E}} f_{vw} \forall v \in \mathcal{V} \setminus (\mathcal{S} \cup \mathcal{I}) \right\}. \quad (5)$$

That is, flow is capacity-feasible and conserved at all *non-terminal* nodes; supplier nodes $\mathcal{S}$ may inject net flow and illegal-buyer nodes $\mathcal{I}$ may absorb net flow. Without loss of generality, we take the flow f to be the edge weight, and for any feasible f , we define the total flow delivered into a particular illegal buyer $i \in \mathcal{I}$ as its inflow

$$F_i(f) = \sum_{u:(u,i) \in \mathcal{E}} f_{ui}. \quad (6)$$

Next, we introduce and define the total supplier-to-illegal-buyer throughput as the multi-source, multi-sink max-flow value over our prescribed supply-chain graph $\mathcal{G}$,

$$F_{\mathcal{S} \rightarrow \mathcal{I}}(\widetilde{W}) = \max_{f \in \mathcal{F}(\widetilde{W})} \sum_{i \in \mathcal{I}} F_i(f) = \max_{f \in \mathcal{F}(\widetilde{W})} \sum_{i \in \mathcal{I}} \sum_{u:(u,i) \in \mathcal{E}} f_{ui}. \quad (7)$$

Detection model and utilities. We adopt a sink-interdiction model: if an illegal buyer $i \in \mathcal{I}$ is inspected, illicit throughput terminating at i is seized/blocked. In addition, a shell company can also be interdicted, resulting in all the flow at that node being set to zero. Under without-replacement inspection, the probability that i is not inspected is $1 - c_{\pi_\theta}(i)$. Thus, for a fixed π_θ , the attacker's expected *successful* illicit throughput under a flow f is then simply stated as:

$$\sum_{i \in \mathcal{I}} (1 - c_{\pi_\theta}(i)) F_i(f). \quad (8)$$

The attacker chooses δ (hence $\widetilde{W}$) to maximize expected successful illicit throughput, subject to optimal routing of flow. Including an optional manipulation-effort penalty $\kappa \geq 0$, the follower utility is

$$U_A(\theta, \delta) = \max_{f \in \mathcal{F}(\widetilde{W})} \sum_{i \in \mathcal{I}} (1 - c_{\pi_\theta}(i)) F_i(f) - \kappa \sum_{(u,v) \in \mathcal{E}} |\delta_{uv}|. \quad (9)$$

In contrast, the enforcement agencies, ie, defenders, seek to minimize successful illicit throughput while accounting for inspection costs. A canonical defender utility is:

$$U_D(\theta, \delta) = - \max_{f \in \mathcal{F}(\widetilde{W})} \sum_{i \in \mathcal{I}} (1 - c_{\pi_\theta}(i)) F_i(f) - \lambda \sum_{v \in \mathcal{V}} c_v(p_\theta) C_v, \quad (10)$$

where $C_v \geq 0$ is a per-node inspection burden and $\lambda \geq 0$ trades off interdiction vs. cost. Alternative utilities can also be considered, such as maximizing the interdiction set I , containing as many shell company and illegal buyer nodes. However, our framing in eq. (10) poses a natural foil to the attacker's utilities.

2.2 Stackelberg equilibrium

Given the definition of the Stackelberg game along with the payoffs of each player, it is natural to consider the notion of equilibria. We can now pose this question explicitly by stating the objective for each player's best response. More precisely, given the leader's commitment π_θ , the follower's best response is

$$\delta^*(\theta) \in \arg \max_{\delta \in \mathcal{D}(W)} U_A(\theta, \delta). \quad (11)$$

The leader then chooses θ anticipating this best response:

$$\theta^* \in \operatorname{argmax}_{\theta} U_D(\theta, \delta^*(\theta)). \quad (12)$$

We refer to this as a *Graph Stackelberg Security Game with Budgeted Edge-Weight Attacks* and a *multi-source, multi-sink max-flow* adversarial objective.

3 Related Work

Security games and Stackelberg models. Stackelberg security games model a defender who commits to a resource allocation strategy and an attacker who best-responds after observing that commitment [Tambe, 2012, Conitzer and Sandholm, 2006]. This framework has supported deployed decision aids in real-world protection settings, demonstrating that game-theoretic allocation can improve outcomes under scarce resources [Fang et al., 2016]. Our work uses the Stackelberg structure but focuses on supply-chain diversion where the attacker adapts by shifting flow through an evolving network.

Network interdiction and bilevel optimization. Network interdiction problems formalize a leader who disables or constrains parts of a network to reduce an adversary’s objective, such as maximum flow [Wood, 1993]. The literature includes algorithmic developments and surveys emphasizing the bilevel nature of interdiction and the computational challenges it creates [Smith and Song, 2020]. Extensions study multi-objective and cost tradeoffs in flow interdiction [Royset and Wood, 2007]. Our formulation aligns with this tradition but emphasizes learning policies that can generalize across changing graphs.

Learning for interdiction and security domains. Recent work explores machine learning and reinforcement learning as scalable alternatives or complements to exact optimization for interdiction-style problems [Baycik, 2022, Huang et al., 2023]. In security-game settings, deep reinforcement learning has been used to compute policies in structured patrol and protection domains under uncertainty and partial observability [Wang et al., 2019].

4 Limitations and Future Work

The framework in this paper is intentionally stylized and subject to several limitations. We assume full observability of the supply-chain network, whereas in practice enforcement agencies operate under substantial uncertainty regarding hidden shell companies, relationships, and routing behavior. Key parameters such as detection probabilities, edge capacities, and adversarial manipulation budgets are estimated rather than empirically calibrated, reflecting limited availability of ground-truth data on successful diversion.

The current formulation evaluates enforcement and adversarial response within a single episode on a fixed graph instance. While the Stackelberg equilibrium conceptually assumes that the leader’s policy is learned or optimized over repeated interactions, we do not model this learning process explicitly. Real-world export control enforcement is inherently repeated, with agencies updating inspection priorities over time and smuggling networks adapting in response. Future work could extend the model to a multi-episode setting, enabling policies to be trained and evaluated across repeated interactions and a distribution of network instances, and allowing analysis of learning dynamics and generalization beyond a single realization.

Similarly, the supply-chain graph is treated as static. In practice, shell companies are created, dissolved, and re-linked in response to enforcement pressure, with routing patterns evolving accordingly. An important extension is to model the supply chain as a dynamic graph with endogenous node and edge evolution, including shell-company birth and death processes, time-varying capacities, and adaptive rewiring. Such dynamics would better capture the lag between reactive designation mechanisms and rapid network reconfiguration.

Finally, scaling the framework to larger or evolving networks motivates learning-based solution methods for settings where exact bilevel optimization becomes intractable. Reinforcement learning approaches combined with graph-based representations offer a natural direction for learning inspection policies that adapt over time while remaining grounded in the Stackelberg structure formalized here.

References

- N. O. Baycik. Machine learning based approaches to solve the maximum flow network interdiction problem. *Computers & Industrial Engineering*, 167:107873, 2022. doi: 10.1016/j.cie.2021.107873. URL <https://doi.org/10.1016/j.cie.2021.107873>.
- Bureau of Industry and Security. Implementation of additional export controls: Certain advanced computing and semiconductor manufacturing items; supercomputer and semiconductor end use; entity list modification. Federal Register, 10 2022. URL <https://www.govinfo.gov/app/details/FR-2022-10-13/2022-21658>. 87 FR 62186.
- Bureau of Industry and Security. Implementation of additional export controls: Certain advanced computing items; supercomputer and semiconductor end use; updates and corrections. Federal Register, 10 2023. URL <https://www.govinfo.gov/app/details/FR-2023-10-25/2023-23055>. 88 FR 73458.
- Bureau of Industry and Security. Know your customer resources, 2025. URL <https://www.bis.gov/media/documents/know-your-customer-resources.pdf>.
- V. Conitzer and T. Sandholm. Computing the optimal strategy to commit to. In *Proceedings of the 7th ACM Conference on Electronic Commerce (EC)*, pages 82–90, 2006. doi: 10.1145/1134707.1134717. URL <https://doi.org/10.1145/1134707.1134717>.
- Executive Office of the President. Adjusting imports of semiconductors, semiconductor manufacturing equipment, and their derivative products into the united states, 2026. URL <https://www.whitehouse.gov/presidential-actions/2026/01/adjusting-imports-of-semiconductors-semiconductor-manufacturing-equipment-and-their-derivative-products-into-the-united-states/>.
- F. Fang, T. H. Nguyen, R. Pickles, W. Y. Lam, G. R. Clements, B. An, R. Singh, M. Tambe, and A. Lemieux. Deploying PAWS: Field optimization of the protection assistant for wildlife security. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2016. URL <https://ojs.aaai.org/index.php/AAI/article/view/19070>.
- E. Grunewald and T. Fist. Countering AI chip smuggling has become a national security priority. Center for a New American Security Working Paper, 6 2025. URL <https://www.cnas.org/publications/reports/countering-ai-chip-smuggling-has-become-a-national-security-priority>.
- D. Huang et al. Solving the shortest path interdiction problem via reinforcement learning. *International Journal of Production Research*, 2023. doi: 10.1080/00207543.2021.2002962. URL <https://doi.org/10.1080/00207543.2021.2002962>. TandF lists 2023 as publication year; DOI corresponds to the article record. Replace author list with full authors as shown in your PDF or the journal page.
- J. O. Royset and R. K. Wood. Solving the bi-objective maximum-flow network-interdiction problem. *INFORMS Journal on Computing*, 19(2):175–184, 2007. doi: 10.1287/ijoc.1060.0191. URL <https://doi.org/10.1287/ijoc.1060.0191>.
- Semiconductor Industry Association. SIA factbook 2025, 2025. URL <https://www.semiconductors.org/wp-content/uploads/2025/05/2025-SIA-Factbook-FINAL-1.pdf>.
- S. Shivakumar, C. Wessner, and T. Howell. The limits of chip export controls in meeting the china challenge. Center for Strategic and International Studies (CSIS) Analysis, 4 2025. URL <https://www.csis.org/analysis/limits-chip-export-controls-meeting-china-challenge>.
- J. C. Smith and Y. Song. A survey of network interdiction models and algorithms. *European Journal of Operational Research*, 283(3):797–811, 2020. doi: 10.1016/j.ejor.2019.06.024. URL <https://doi.org/10.1016/j.ejor.2019.06.024>.
- K. M. Sutter. U.s. export controls and china: Advanced semiconductors. Technical Report R48642, Congressional Research Service, 9 2025. URL <https://www.congress.gov/crs-product/R48642>.

- M. Tambe. *Security and Game Theory: Algorithms, Deployed Systems, Lessons Learned*. Cambridge University Press, 2012. doi: 10.1017/CBO9780511973031. URL <https://doi.org/10.1017/CBO9780511973031>.
- U.S. Department of Justice, U.S. Attorney’s Office, Middle District of Florida. U.s. citizens and chinese nationals arrested for exporting artificial intelligence technology in violation of u.s. export controls, 11 2025. URL <https://www.justice.gov/usao-mdfl/pr/us-citizens-and-chinese-nationals-arrested-exporting-artificial-intelligence-technology>. Press release, November 20, 2025. Hon Ning Ho, Cham Li, and others were indicted in connection with an alleged export control smuggling scheme using Janford Realtor LLC.
- U.S. Department of Justice, U.S. Attorney’s Office, Southern District of Texas. U.s. authorities shut down major china-linked ai tech smuggling network, 12 2025. URL <https://www.justice.gov/usao-sdtx/pr/us-authorities-shut-down-major-china-linked-ai-tech-smuggling-network>. Press release, December 8, 2025. Alan Hao Hsu and others pleaded guilty to smuggling export-controlled Nvidia GPUs in violation of U.S. export laws.
- Y. Wang, Z. R. Shi, B. An, et al. Deep reinforcement learning for green security games with real-time information. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2019. doi: 10.1609/aaai.v33i01.33011401. URL <https://doi.org/10.1609/aaai.v33i01.33011401>.
- R. K. Wood. Deterministic network interdiction. *Mathematical and Computer Modelling*, 17(2):1–18, 1993. doi: 10.1016/0895-7177(93)90236-R. URL [https://doi.org/10.1016/0895-7177\(93\)90236-R](https://doi.org/10.1016/0895-7177(93)90236-R).